Google Cloud H2S

# Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

**Participant Details**

a. Participant Name: Aditya Hegde (adityahegde007@gmail.com)
b. Problem Statement: Post-surgery recovery workflows are fragmented across disconnected notes, scheduling tools, and manual follow-ups. VitalFlow unifies this into a multi-agent productivity assistant that coordinates clinical and logistics actions in real time.

## Brief about the idea

VitalFlow is a multi-agent productivity assistant for post-surgery recovery operations.

Core Concept: A Primary Orchestrator receives user intent and delegates to Clinical Analyst and Logistics Officer sub-agents.

Innovation: Realtime orchestration + tool execution + database memory for reliable multi-step workflow completion.

# Your solution should be able to explain the following:

- *How did you approach the common problem statement and translate it into a working solution using ADK, MCP/AlloyDB AI while ensuring Hackathon's core requirements are met?*
- *What real-world problem does your solution address, and what practical impact does it create for users?*
- *What is the core approach or workflow behind your solution?*

- We built an API-first orchestration system where a backend primary agent coordinates sub-agents and tools in sequence.
- Core Workflow: User intent -> Orchestrator -> Clinical reasoning/history -> Logistics scheduling -> DB updates -> Realtime UI updates.
- Impact: Reduces coordination delays and turns multi-step recovery operations into a single guided interaction.

# Opportunities

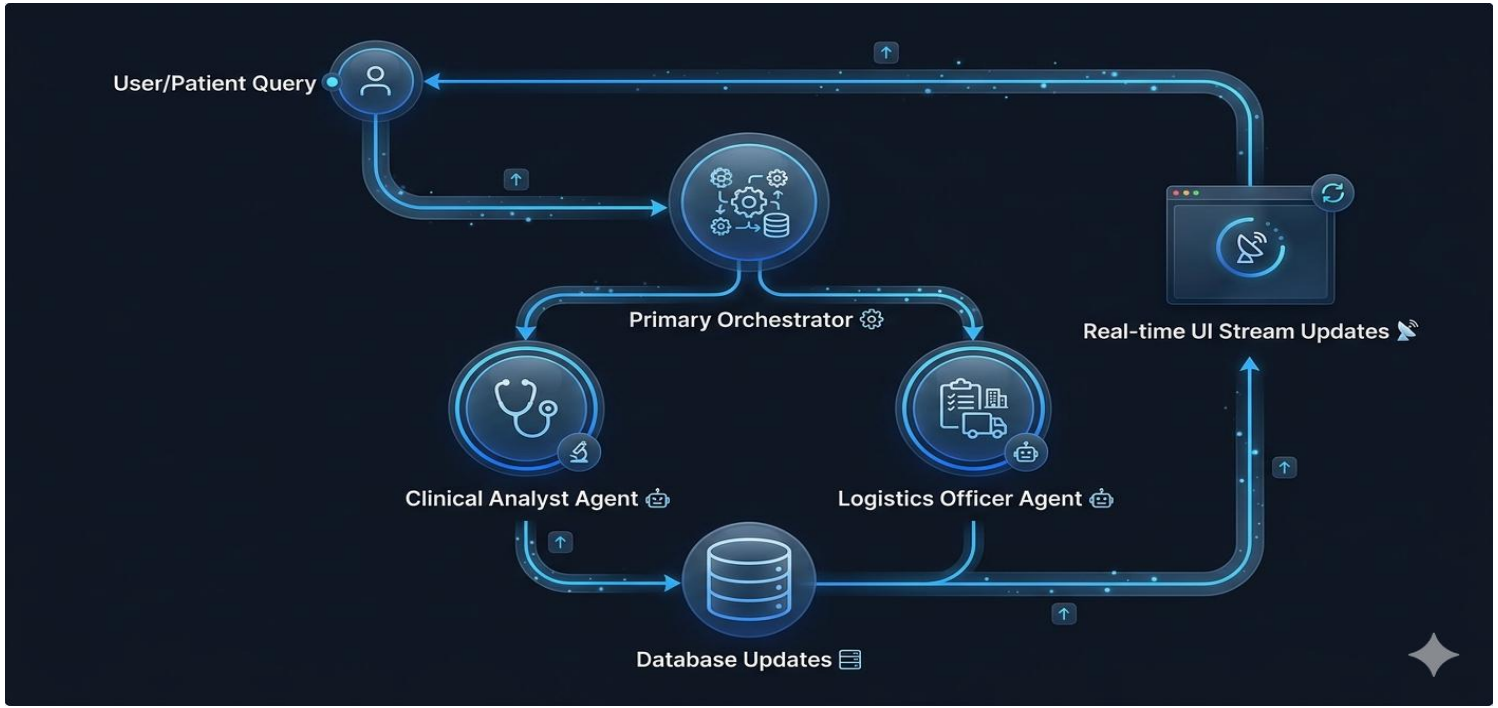
Differentiation: VitalFlow executes coordinated actions, not just static dashboard views.

USP: Transparent real-time trace of agent actions, tool usage, and outcomes in audit streams and run-event timelines.

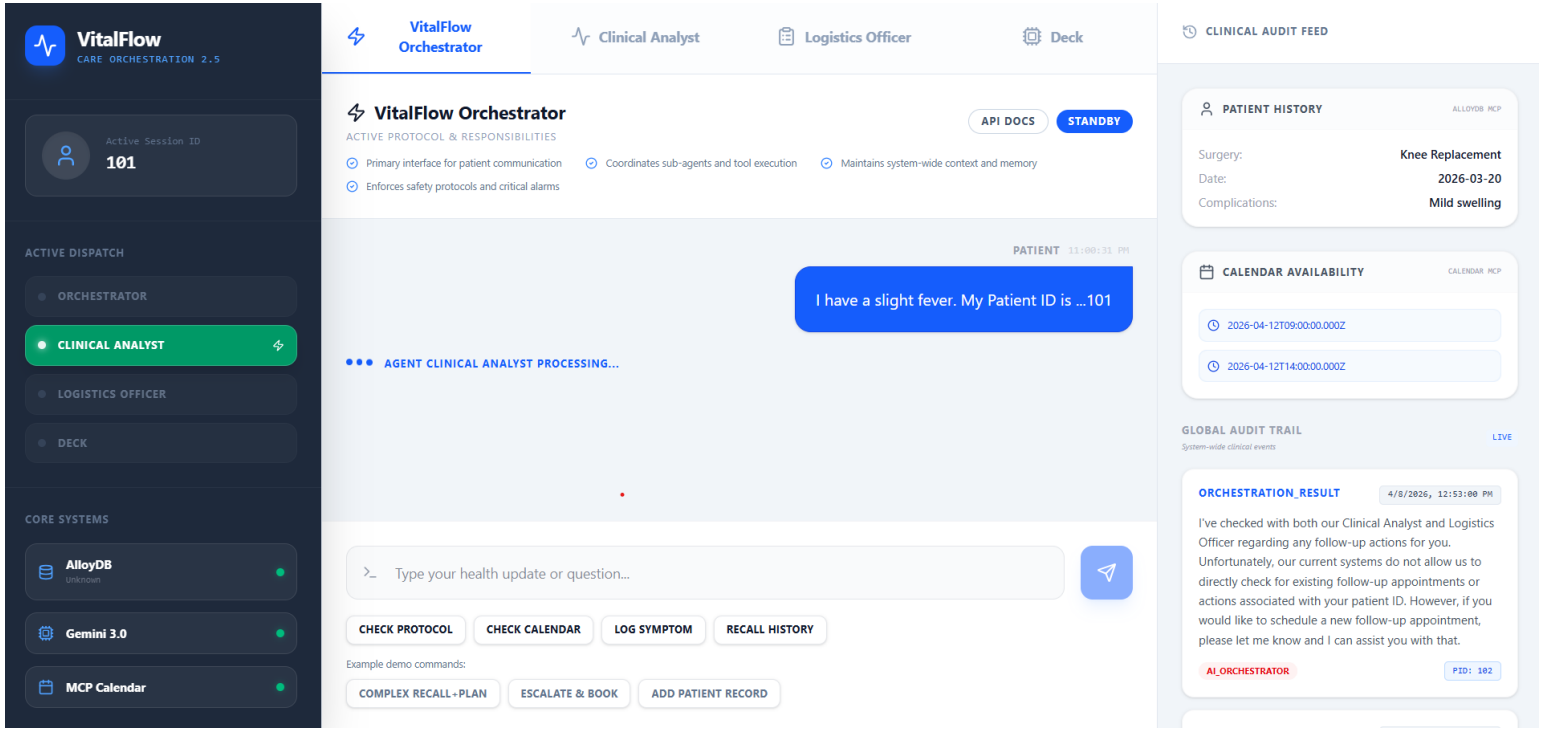
## List of features offered by the solution

- Primary Orchestrator with Clinical + Logistics sub-agents
- Realtime run events (SSE)
- Realtime audit trail + calendar updates
- Patient history read/write APIs
- Dynamic slot availability and booking
- Structured DB persistence (PostgreSQL/SQLite)
- MCP-compatible tool registry and tool invocation
- OpenAPI + Swagger docs for API-based deployment

# Process flow diagram or Use-case diagram



# Wireframes/Mock diagrams of the proposed solution

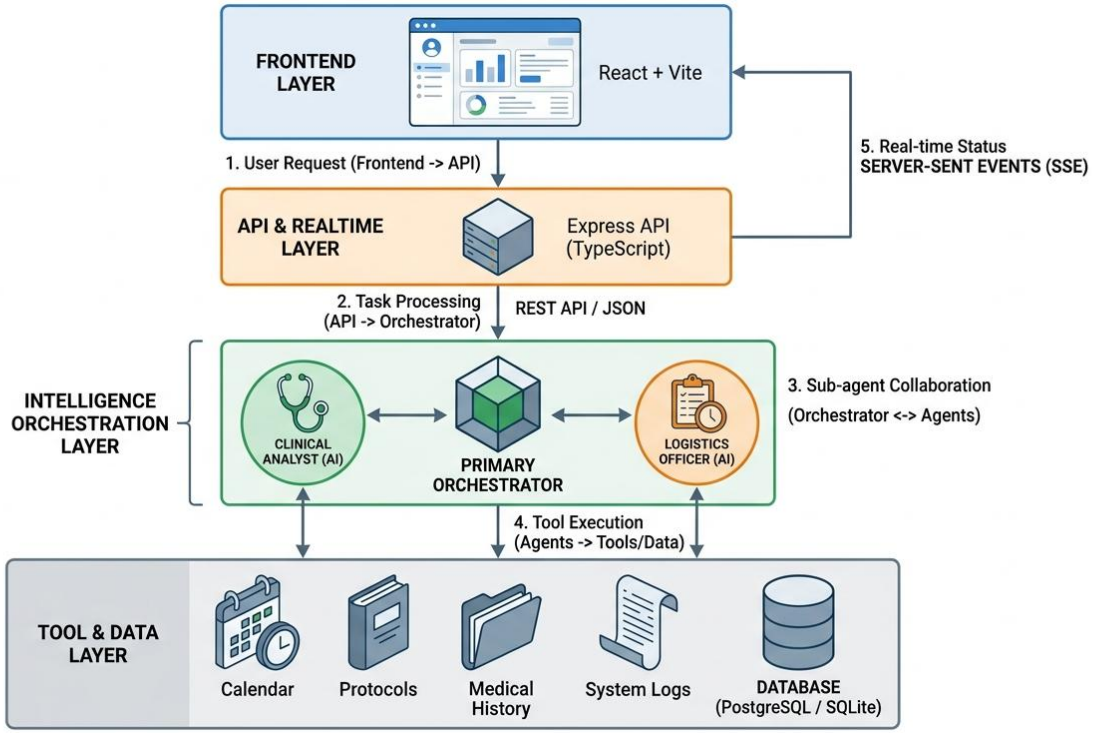



1) Left: Patient Identity + Active Dispatch + Core Systems

2) Center: Agent conversation + orchestration flow

3) Right: Patient history + protocol + calendar availability + audit trail

# Architecture diagram of the proposed solution:

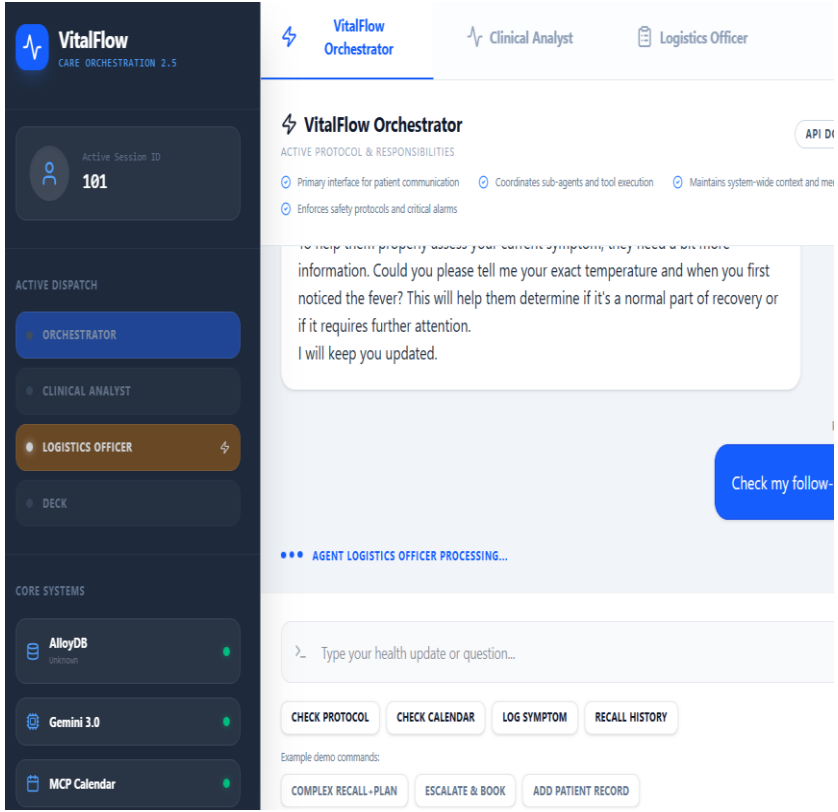


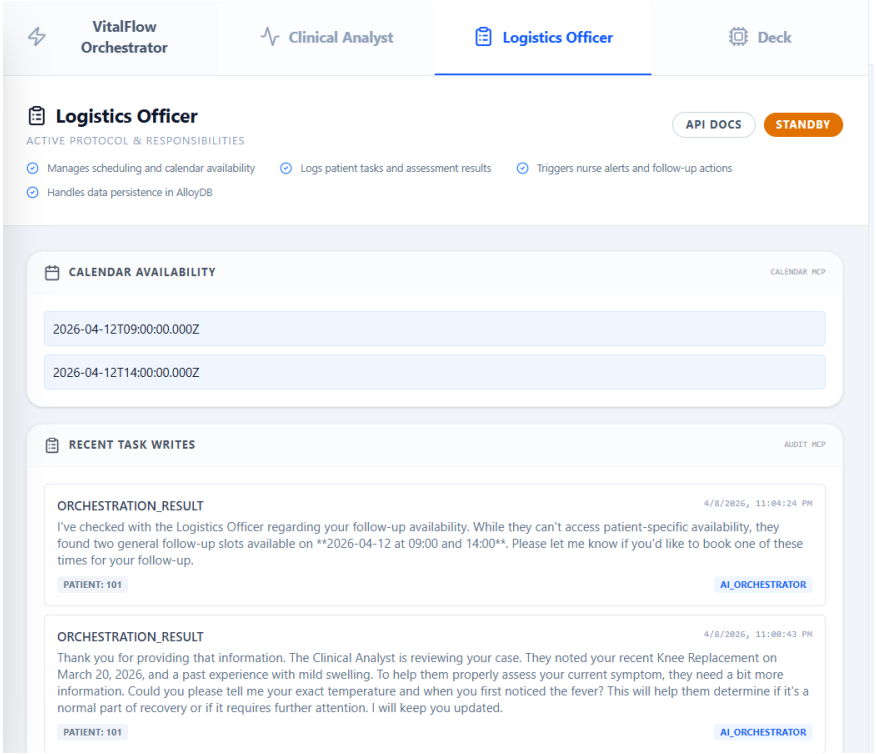
# Technologies/Google Services used in the solution

- Gemini API for orchestration reasoning
- Node.js + Express + TypeScript backend
- React + Vite frontend- PostgreSQL (pg) and SQLite
- Dockerized Cloud Run deployment
- OpenAPI + Swagger for API discoverability

Why this stack: fast development, scalable API-first architecture, and real-time agent observability.

# Snapshots of the prototype



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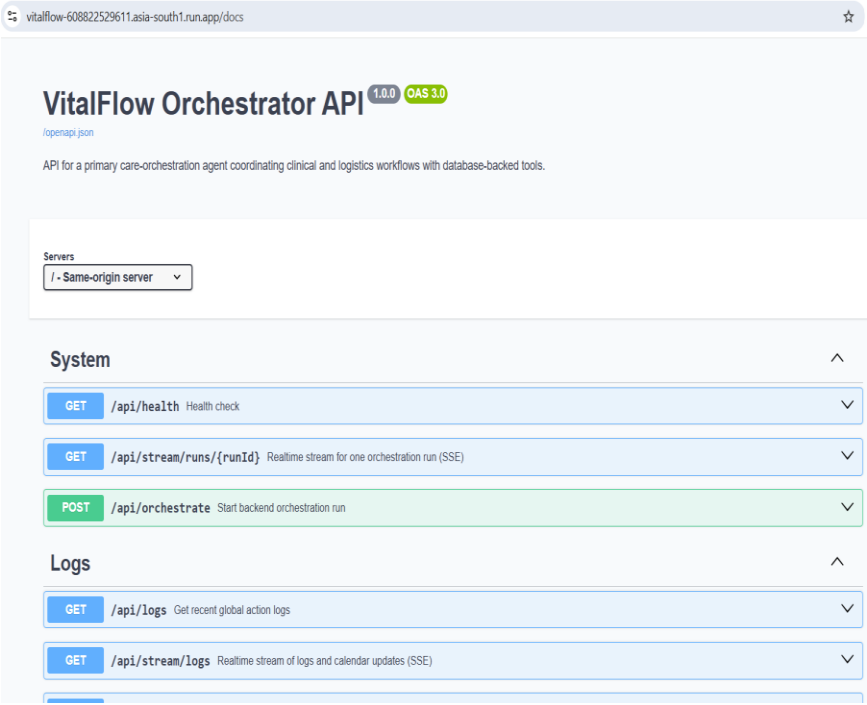


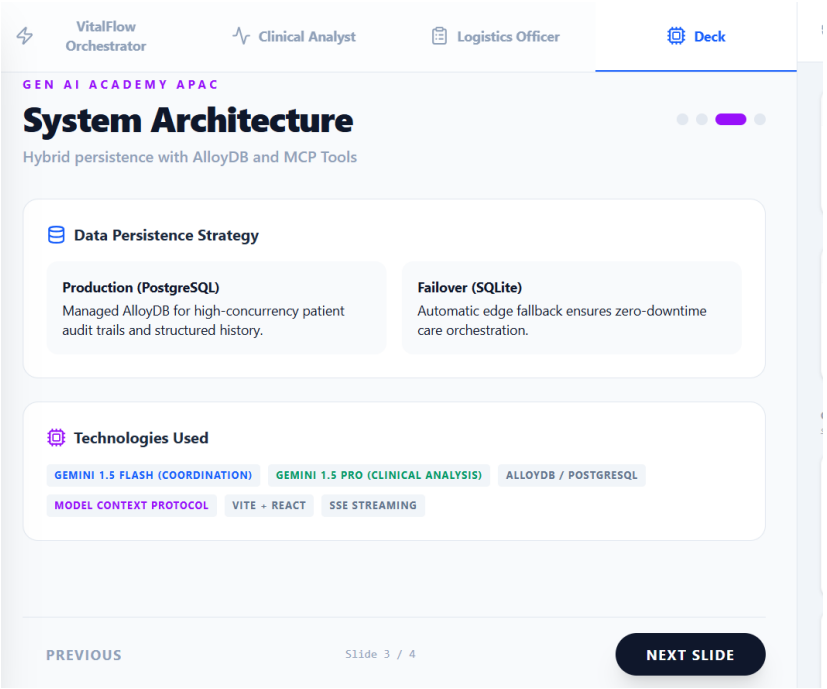

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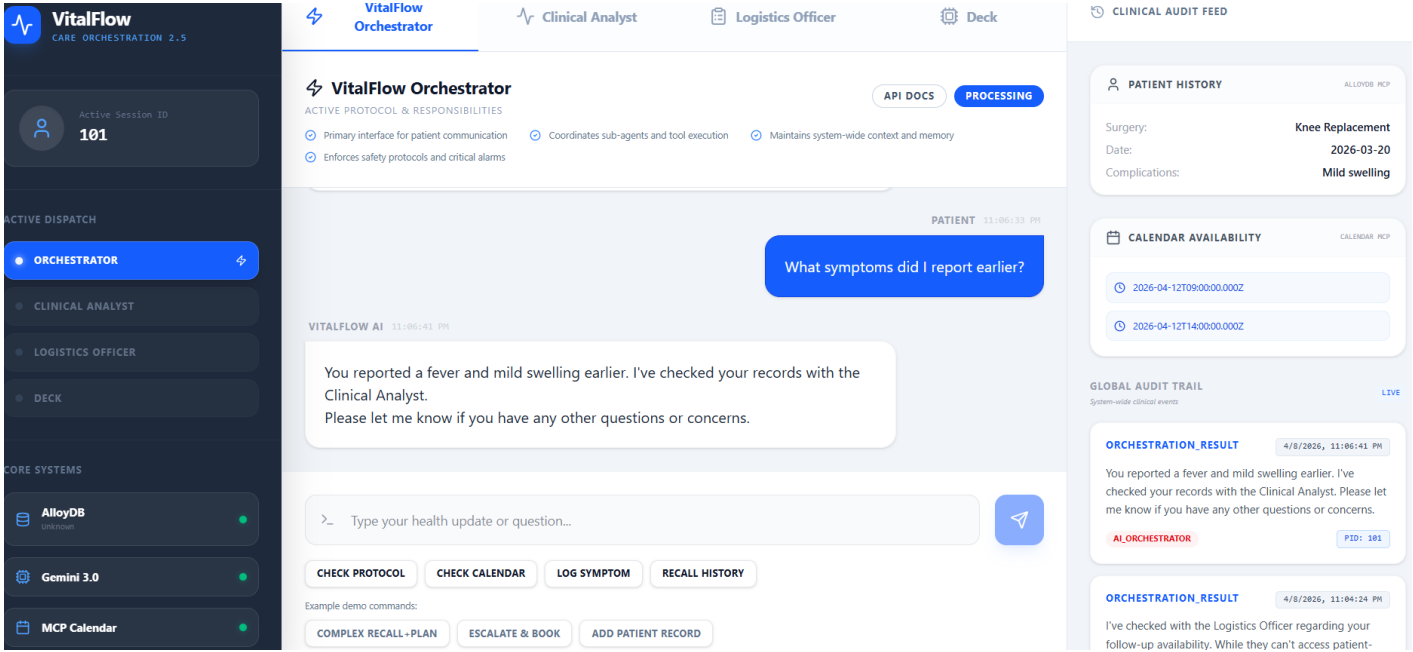

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# Thank you